# Muhammad Noman Ashraf, PhD

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## Short Bio

Ph.D. Electrical Engineer specializing in **power electronics, grid-connected converters, nonlinear control, and microgrid stability**, supported by **three years of industrial R&D experience in South Korea**. My research focuses on **power quality enhancement, synchronization under weak and distorted grid conditions, harmonic compensation, and inter-converter oscillation mitigation** in converter-dominated renewable energy systems. I have **hands-on experience in the hardware design and development of inverter systems and bidirectional DC–DC converters**, including converter prototyping, PCB design, DSP-based digital control implementation, and experimental validation.

## Education

### Ph.D. in Electrical Engineering — August 25, 2022–May 19, 2026
**Khalifa University, Abu Dhabi, UAE** — CGPA: 3.68/4.0

- **Dissertation Title:** *Power Quality Enhancement for Grid Connected Inverters in Renewable Energy Systems*

### MSc in Electrical Engineering — September 1, 2017–August 16, 2019
**Soongsil University, Seoul, South Korea** — CGPA: 4.39/4.5

- **Master's Thesis:** *Digital Lock-In Amplifier Based Harmonic Compensation and Islanding Detection for Single-Phase Grid-Connected Inverter Systems*

### B.E. in Electronics — August 27, 2012–August 7, 2016
**Karachi Institute of Economics and Technology (KIET), Karachi, Pakistan** — CGPA: 3.7/4.0

## Industrial and Research Employment Experience

### Research and Hardware Development Engineer — August 2019–July 2022
**OKY Ltd., Seoul, South Korea**

- Led the complete design and testing of a **10 kW dual-stage bidirectional DC–DC–AC converter**.
- Performed topology verification in PSIM and handled circuit, inductor, transformer, and PCB design.
- Selected power components for prototyping and carried out complete hardware/software implementation using TI DSP platforms.
- Conducted experimental validation with bidirectional power flow.
- Developed software and control algorithms for **single-phase inverter-based dynamic voltage compensators (DVCs)**.
- Designed supercapacitor charging algorithms and fast voltage sag detection/compensation schemes.
- Achieved smooth transition of load from grid to inverter within approximately 2 ms.

- Managed firmware and hardware revision histories for commercial single-phase products.
- Contributed to new project proposal writing, including topology analysis, loss estimation, rating calculations, and preliminary cost estimation.

**Research Assistant** **Jun 2016–Aug 2016**
**NED University of Science and Technology, Karachi, Pakistan**

- Worked in CIS Robotics Lab on vibrotactile feedback for 3D printed prosthetic hands.
- Programmed the controller and interfaced prosthetic sensors with the embedded platform.
- Contributed to work published at IEEE ICRAI 2016 and accepted at Asia Haptics Japan 2016.

## Publications

### IEEE Transactions

[1] M. N. Ashraf, M. S. E. Moursi, K. Al Hosani, and A. Al Durra, "Andronov–Hopf Oscillator Based Harmonic Elimination Scheme for Grid-Connected Inverters," *IEEE Transactions on Smart Grid*, Early access.

[2] M. N. Ashraf, M. S. E. Moursi, K. Al Hosani, and A. Al Durra, "Andronov–Hopf Oscillator-Based Phase-Locked Loop for Three-Phase Grid Connected Inverters," *IEEE Transactions on Power Electronics*, vol. 40, no. 6, pp. 7634–7639, June 2025, doi: 10.1109/TPEL.2025.3531901.

[3] M. N. Ashraf, R. A. Khan, and W. Choi, "A Novel Selective Harmonic Compensation Method for Single-Phase Grid-Connected Inverters," *IEEE Transactions on Industrial Electronics*, vol. 68, no. 6, pp. 4848–4858, June 2021, doi: 10.1109/TIE.2020.2989723.

[4] Mamdouh, et al., "Novel Control Strategy for Hydrogen Electrolyzer to Ride-Through Coupled Frequency and Voltage Disturbance Events," *IEEE Transactions on Industry Applications*, Early Access.

[5] M. N. Ashraf, M. S. E. Moursi, K. Al Hosani, and A. Al Durra, "Adaptive Energy-Based Grid-Forming Control for Mitigating Interconverter Oscillations," *IEEE Transactions on Industry Applications*, **Major Revision R1**.

[6] M. N. Ashraf, M. S. E. Moursi, K. Al Hosani, and A. Al Durra, "Latency-Constrained Intrusion Detection Co-Designed with Fast Dynamics of Grid-Forming Converters," *IEEE Transactions on Power Electronics*, **Submitted and Under Review**.

[7] Abdullahi, et al., "Estimation of Time-Varying Inertia Based on Power System States Using Adaptive Multitask Transformer-BiLSTM with Dynamic Attention and Uncertainty Aware Loss" *IEEE Transactions on Smart Grid*, **Submitted and Under Review**.

[8] Biyadgie, et al., "Transient Stability Improvement for Grid Forming Converter Operating as DC Voltage Controller in HVDC System" *IEEE Transactions on Power Electronics*, **Submitted and Under Review**.

### Other Q1 Journals and Conferences

[9] M. N. Ashraf, R. A. Khan, A. S. Akram, et al., "A novel PLL technique using digital lock-in amplifier under distorted grid conditions," *Journal of Power Electronics*, vol. 26, pp. 818–830, 2026.

[10] M. N. Ashraf, A. S. Akram, and W. Choi, "Novel Hybrid Islanding Detection Technique Based on Digital Lock-In Amplifier," *Energies*, vol. 18, 3449, 2025.

[11] R. A. Khan, M. N. Ashraf, and W. Choi, "A Harmonic Compensation Method Using a Lock-In Amplifier under Non-Sinusoidal Grid Conditions for Single Phase Grid Connected Inverters," *Energies*, vol. 14, 597, 2021.

[12] S. Amin, M. N. Ashraf, and W. Choi, “A New Harmonic Compensation Method...” *Journal of Power Electronics*, vol. 25, no. 5, pp. 358–368, 2020.

[13] Muhammad Noman Ashraf and Woojin Choi, “A Novel Hybrid Islanding Detection Method Using Digital Lock-In Amplifier,” *Annual Summer Conference of KIPE*, South Korea, June 2019.

[14] Muhammad Noman Ashraf, Reyyan Ahmed Khan, and Woojin Choi, “A Novel Harmonic Compensation Technique for the Grid-Connected Inverters,” *Annual Summer Conference of KIPE*, South Korea, June 2019.

[15] Muhammad Noman Ashraf, Reyyan Ahmad Khan, and Woojin Choi, “A Robust PLL Technique using a Digital Lock-In Amplifier under the Non-Sinusoidal Grid Conditions,” *ICPE 2019 – ECCE Asia*, Busan, South Korea, May 2019.

[16] Muhammad Noman Ashraf, Reyyan Ahmed Khan, and Woojin Choi, “A Robust PLL Technique using a Digital Lock-In Amplifier under the Non-Sinusoidal Grid Conditions,” *Annual Winter Conference of KIPE*, South Korea, November 2018.

[17] Reyyan Ahmad Khan, Muhammad Noman Ashraf, and Woojin Choi, “A Robust Harmonic Compensation Technique using Digital Lock-In Amplifier...” *ICPE 2019 – ECCE Asia*, Busan, South Korea, 2019.

[18] Reyyan Ahmad Khan, Muhammad Noman Ashraf, and Woojin Choi, “A Novel Harmonic Compensation Method for the Single-Phase Grid Connected Inverters,” *Annual Summer Conference of KIPE*, Gangwon, South Korea, July 2018. **Best Conference Paper Award**.

[19] Reyyan Ahmad Khan, Muhammad Noman Ashraf, and Woojin Choi, “A Method to Improve the Performance of Phase-Locked Loop (PLL)...” *Annual Winter Conference of KIPE*, Seoul, South Korea, November 2017. **Best Conference Paper Award**.

## Teaching Experience

### Teaching Assistant — August 2022–May 2026

**Khalifa University, Abu Dhabi, UAE**

- Teaching undergraduate laboratory courses for **Communication Systems**, **Instrumentation and Measurement**, and **Electric Circuits**.
- Delivered pre-laboratory lectures to explain experimental objectives, theoretical background, and measurement procedures.
- Guided undergraduate students during laboratory experiments, troubleshooting, data acquisition, and result interpretation.
- Supported students in understanding circuit behavior, instrumentation usage, and experimental verification.

### Lab Engineer/Instructor — Jan 2017–Sep 2017

**KIET, Karachi, Pakistan**

- Served as teaching assistant for Instrumentation and Measurement Laboratory.
- Conducted undergraduate laboratory instruction and practical demonstrations.
- Supported simulation and design activities in LabVIEW and MATLAB.
- Assisted with sensor interfacing and hardware debugging.

## Technical Skills

- **Power Electronics:** Single/three-phase inverters; isolated/non-isolated DC–DC converters; hybrid AC/DC grids.
- **Control and Modeling:** PLL design; harmonic compensation; grid-following and grid-forming inverter control; current and voltage controller design; weak-grid synchronization; nonlinear oscillator-based methods; energy-based control.
- **Hardware and Embedded Systems:** TI DSPs; dSPACE, OPAL-RT, PCB design; power circuit design; PIC microcontrollers; embedded C/C++; prototyping and experimental validation.
- **Software Tools:** MATLAB; PSIM; OrCAD; KiCAD; LaTeX.
- **Other Skills:** Project Proposal Writings, EHS documentations, risk assessments for Laboratories.

## Awards and Achievements

- International Graduate Research Scholarship, Soongsil University, South Korea.
- University Merit Scholarship for Engineering Studies for Undergraduate.
- Ranked among the top 5 engineering students during Undergraduate studies.
- Best Conference Paper Awards at KIPE conferences (South Korea).
- Dean's Honours Badge in multiple semesters during Undergraduate studies.
- President's Honours Badge in multiple semesters during Undergraduate studies.
- First position as a volunteer organizer at debate competitions during Undergraduate studies.

## References

**Prof. Mohamed Shawky El Moursi, IEEE Fellow**
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Khalifa University, Abu Dhabi, UAE
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**Prof. Woojin Choi**
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